

From Scores to Kinds: Projectibility and Validity in AI Evaluation

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3rd July 2026

Abstract

AI evaluation terms let scores and labels travel beyond the cases that produced them. A benchmark result becomes evidence for REASONING; an attack result becomes evidence of JAILBREAK RESISTANCE; an aggregate label becomes a governance claim about ALIGNMENT or TRUSTWORTHINESS. Such terms function as inference licences: they authorize movement from an observation or score to a claim about recurrence, vulnerability, capability, or governance. Evaluation practice has explicit norms for two questions. Reliability asks whether a result is stable; construct validity asks what it means in the measured setting. Neither says when the licensed inference may travel, or what keeps it alive.

This paper supplies the structure of that third norm. Drawing on Goodman's projectibility and on Boyd's and Khalidi's accounts of kinds, I treat each particular evaluation as issuing a licence with a life-cycle. A licence is generated under validity conditions, or blocked, or vitiated (contamination is the clearest vitiator); it's bounded in scope; and once live it can be narrowed by new evidence, eroded by optimization against the measure, defeated by stronger results, revoked, or left to expire as models and policies change. Two axes cut across every stage: misrecognition, when uptake outruns what was licensed, and maintenance, since stability under repeated measurement isn't control. A four-role typology of evaluation terms falls out as a diagnostic, and worked cases (hallucination; reasoning and emergence; jailbreak; alignment and trustworthiness, with sycophancy as pressure case) show the life-cycle sorting disputes the labels alone can't: the emergence debate, for one, becomes a definite question about when a capability licence forms.

Keywords: AI evaluation; benchmarks; construct validity; projectibility; inference licences; natural kinds

I INTRODUCTION

AI evaluation often moves from a score to a much larger claim. A benchmark result is reported as evidence for REASONING. A safety evaluation says that a system has JAILBREAK RESISTANCE. A

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model card, audit, or policy document describes an assistant in terms of ALIGNMENT, TRUSTWORTHINESS, or HALLUCINATION risk. These labels point to observed behaviour and invite readers to draw inferences: that a failure will recur, that a defence will generalize, that a capability will transfer, or that a system is suitable for some governance purpose.

The debate over emergent abilities in large language models shows the problem in a compact form. Wei et al. (2022) argue that some abilities appear only once models reach sufficient scale. Schaeffer et al. (2023) reply that some apparent jumps can be produced by the choice of metric: discontinuous scoring can make gradual changes in performance look like sudden emergence.

The dispute reaches beyond the acquisition of new abilities. It asks what a score licenses us to infer. If a benchmark curve has a sharp bend, can we infer a new capability, a change in the measurement instrument, or something more local about a task, metric, and model family?

The problem is general: Raji et al. (2021) argue that influential benchmarks are often treated as stand-ins for broad claims about general ability, even though the benchmarks themselves are finite, contextual, and constructed around specific tasks. A benchmark can be useful while still being too narrow to support the claim attached to it. AI evaluation terms are vague in places, but the deeper issue is inferential. They function as INFERENCE LICENCES: they authorize a move from an observation or score to a claim about recurrence, vulnerability, capability, or governance. The idea has a pedigree. Ryle treated law-like statements as inference tickets, statements whose content is what they entitle us to infer rather than a hidden state they describe (Ryle, 1949). Different evaluation terms authorize different moves.

This paper asks when such licences are earned, how far they travel, and what keeps them alive. A term licenses a type of move; a particular evaluation issues a particular licence; and the licences, not the terms, are what have careers. Which move each term licenses is the diagnostic sub-question, and the worked cases pursue it term by term.

I use PROJECTIBILITY in a plain sense first: a classification is projectible when applying it in one case gives us a warranted reason to expect something in another. The term is Goodman's, coined for the predicates that sustain induction (Goodman, 1955); I extend it to the classifications an evaluation practice deploys.

If a system produces one unsupported answer, that alone doesn't show a general tendency to hallucinate. If an attack succeeds once, that alone doesn't show that the model is generally unsafe. If a model passes one reasoning benchmark, that alone doesn't show a stable reasoning capability. The classification becomes useful when we can say what else should follow from it, in which domain, and under which limits.

In a classical definition, a term gives necessary and sufficient conditions for membership. Boyd's account of HOMEOSTATIC PROPERTY CLUSTERS and Khalidi's causal-network account show how classes can support projection without that form of definition (Boyd, 1991, 1999; Khalidi, 2018). For AI evaluation, the lesson is modest: useful terms needn't have essences, but they should mark structures that support defensible inferences beyond the immediate observation.

The CONSTRUCT VALIDITY tradition gives the measurement side of the same problem. A test score doesn't interpret itself. It needs a construct, an operationalization, a metric, a population of instances, and an intended use. Kane's argument-based view of validation is useful here because it treats validity as a question about the interpretations and uses licensed by scores (Kane, 2013). AI evaluations should be read the same way. A label such as HALLUCINATION or JAILBREAK belongs

to a measurement practice that includes tasks, prompts, raters or automated judges, policies, metrics, populations, and deployment decisions.

The division of labour is the paper’s hinge, and it has three parts rather than the familiar two. Reliability asks whether a result is stable when the measurement is repeated. Construct validity asks what a score or label licenses in the measured setting. Projectibility asks when that licensed inference can travel to another model, benchmark, attack, or deployment, and what maintains it over time. Evaluation practice has explicit norms and methods for the first two questions. For the third it mostly has habits.

Neither the kind theory nor the validity theory is the contribution claimed here. Boyd and Khalidi supply projection without essences; Kane supplies the argument-based view of validation; Goodman supplies the vocabulary. The contribution is a structure none of them supplies: a life-cycle for evaluation licences. A licence is generated under stated conditions, or blocked, or vitiated; it’s bounded in scope; and once live it can be narrowed, eroded, defeated, or revoked, or left to expire, while the community’s uptake tracks that career accurately or misreads it. The life-cycle turns projectibility from a compliment paid to good benchmarks into a set of stages with stateable norms.

The argument proceeds as follows. Section 2 presents evaluation terms as inference licences and connects them to construct validity. Section 3 introduces projectibility without essences and distinguishes it from formal transportability. Section 4 sets out the licence life-cycle and restates the paper’s typology of evaluation terms as a diagnostic that falls out of it. Sections 5 and 6 work through the two central cases, hallucination and capability attribution; the second returns to the emergence dispute and locates it at a definite stage of the life-cycle rather than adjudicating it. Section 7 treats jailbreak more briefly, as the case where licences age fastest. Section 8 recasts alignment and trustworthiness as thin verdicts issued by coordination regimes, with sycophancy as the pressure case. Section 9 answers objections and concedes limits.

2 EVALUATION TERMS AS INFERENCE LICENCES

Evaluation language climbs an inference ladder. A raw output, interaction, or score becomes an evaluation label. The label receives a construct interpretation. The construct supports a kind attribution. The attribution then enters a model comparison, release decision, procurement standard, or governance claim. Each step can be legitimate, but the warrant changes as the claim climbs.

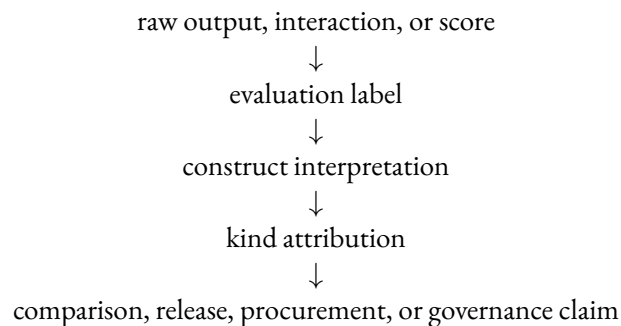


Figure 1: The inference ladder in AI evaluation.

Consider a mathematical-reasoning benchmark. Raw completions are first coded as right or wrong on individual items. The metric aggregates those judgements into an accuracy score. The

benchmark report then interprets the score as evidence for a construct such as MATHEMATICAL REASONING. A leaderboard turns that interpretation into a model ranking, and a user can then treat the ranking as evidence for deployment suitability. The same number appears throughout, but the claim has changed at every rung.

The narrow inference can be sound while the wider one fails. A high score on one population of word problems can license a claim about performance on those items, under that prompt format and scoring rule. It doesn't yet license a general reasoning claim across paraphrases, irrelevant clauses, new item generators, languages, tool scaffolds, or contamination controls. The ladder makes the analytic question visible: which movement is being licensed, and what evidence supports it?

Construct validity names the first warrant. A score has meaning only relative to a construct, an operationalization, a metric, a population of instances, and an intended use (Chouldechova et al., 2024; Kane, 2013). An operationalization is the concrete test procedure that makes a construct measurable. Here, validity is local: the relevant warrant connects a score to a stated interpretation and use. Work on LLM benchmarks makes the same point in the evaluation setting: a benchmark operationalizes a phenomenon through tasks and metrics, then supports a claim only if those parts line up (Bean et al., 2025; Freiesleben, 2026).

In LLM evaluation, repeated probing often makes RELIABILITY easier to estimate: evaluators can ask whether results are stable across item samples, prompt variants, decoding settings, and evaluator systems. Reliability sharpens the local warrant, but it doesn't settle interpretation. The validity question is which probes are compressed into which score, for which construct interpretation and use.

Model comparison makes that compression consequential. Ranking model A above model B requires more than running the same benchmark twice. It requires a judgement that the chosen score summarizes the right profile of tasks, prompts, languages, evaluator systems, and deployment conditions for the intended comparison. A leaderboard can be valid as a narrow benchmark comparison and fail as a governance conclusion if those supports don't hold (Raji et al., 2021).

A licence isn't the same as its uptake. When a leaderboard, a model card, or a procurement office treats a score as supporting a claim, that's uptake: real, consequential, and sometimes wrong. What the score licenses is fixed by its generation conditions and scope, not by how confidently it's repeated. Mistaken uptake has effects (deployments, purchases, headlines), but it doesn't widen the licence, any more than an official record of a permission widens the permission. The gap between licence and uptake isn't noise. It's one of the framework's objects of study, and Section 4 gives it a name.

Model comparison also marks the limit of construct validity. Validity can say what the score licenses here, but it doesn't settle whether the claim travels to another model family, attack distribution, user population, or deployment. That's the projectibility question.

3 PROJECTIBILITY WITHOUT ESSENCES

The projectibility question turns on structure rather than classical definitions. A classical definition gives necessary and sufficient conditions for class membership. The accounts I draw on reject that requirement. Boyd's account of HOMEOSTATIC PROPERTY CLUSTERS explains how a class can support projection when its properties cluster through stabilizing mechanisms. Khalidi's causal-network account explains projectibility through relations among properties, where some properties are more central and help explain the presence or stability of others (Boyd, 1991, 1999; Khalidi, 2018).

Here, `CLASS` and `KIND` do different jobs. A class is any grouping introduced by an evaluation practice. A kind, in the stronger sense used here, is a class whose structure supports projection. The paper doesn't ask whether every label is a kind. It asks when an evaluation class earns that stronger status. The stipulation isn't circular, because the order of analysis runs from structure to projection: the worked cases identify the structure first and derive the licensed inferences from it.

A classification is projectible when applying it in one setting gives a warranted reason to expect something in another setting, under stated conditions. The conditions matter. Projection is field-relative: what travels depends on the domain, purpose, population, and use of the evaluation. A label can project across nearby prompt variants while failing to project across languages, tool access, deployment settings, or adversary classes.

Suppose a red-team test finds that an adversarial suffix (a string appended to a prompt) makes a chat assistant answer a prohibited request. The result supports a narrow projection. If the suffix still works after a few tokens are swapped, after a harmless phrase is added before it, or when another optimizer seed produces a similar appended string, the test gives reason to expect similar vulnerability under the same chat interface, refusal policy, and evaluator standard.

The same result says little about an indirect prompt-injection case where a browser agent reads a web page that tells it to send a private file. The attacker, channel, policy boundary, and system boundary have changed, so the original classification no longer carries the same expectation.

In AI evaluation, projectibility-relevant structure is artificial, engineered, and institutional. It can include training data, model architecture, training objectives, post-training incentives, decoding procedures, retrieval design, prompt format, interface constraints, policy boundaries, evaluator models, benchmark construction, reporting conventions, and deployment use. None of these is necessary or sufficient; none is essential. Together, though, they can explain why a failure profile, vulnerability, capability attribution, or governance bundle recurs, and why projection weakens when those supports change.

The travel question already has a formal treatment. Bareinboim and Pearl's transportability results give graphical conditions under which causal and statistical relations estimated in one setting carry to another (Bareinboim & Pearl, 2016). The formalism presupposes two things the evaluation case usually lacks. The variables are fixed in advance, and the differences between source and target are encoded in a causal model over those variables. Projectibility, as used here, is the prior question: which classifications deserve to be treated as variables at all. That's Goodman's question, not a transport question, and no one can write a selection diagram over incentive structure, refusal machinery, and evaluator standard. Transport is also silent about use: nothing in it distinguishes the same score feeding a release gate from the same score feeding a procurement rule. Both gaps matter for evaluation, and both are where the kind-theoretic accounts do their work.

A structure matters for projectibility when it does two jobs: it explains why an inference travels and where it stops. A resampling detector gives the simple case. The detector treats variation across sampled answers as evidence that a claim is unsupported. The signal can travel to settings where unsupported claims vary across samples. It travels poorly to a model that repeats the same unsupported answer every time, because the signal has disappeared.

Structure of this sort explains travel at a time. But licences also live in time: they're issued, worn down, overtaken, and withdrawn. The next section sets out that life-cycle, and the worked cases then apply both questions together: what structure lets an inference travel, and what stage of its career the

licence has reached.

4 THE LIFE-CYCLE OF AN EVALUATION LICENCE

An inference licence isn't issued once and for all. It's generated under conditions, bounded in scope, and then kept, narrowed, worn out, overtaken, or withdrawn. The shape is familiar from everyday normative statuses such as permissions, which are granted under conditions, bounded, and later revoked, exhausted, or exceeded. Nothing metaphysical rides on the analogy. Every stage below names something evaluation practice already does; what the life-cycle adds is order, assigning each failure its own name instead of the one word *invalid*.

4.1 GENERATION AND SCOPE

A licence is generated when the validity argument goes through: a construct, an operationalization, a metric, an item population, and an intended use that line up (Bean et al., 2025; Kane, 2013). Generation has three outcomes, not two.

In the central case the licence forms. The evaluation supports its stated interpretation, and the licence's scope is fixed by what was actually sampled and held constant: the item population, prompt format, evaluator standard, system boundary, and time of measurement.

Sometimes nothing forms. Benchmark contamination is the clearest case: if the test items sat in the training data, a perfect score licenses nothing about the capability, because what was measured is closer to retrieval of the answer key. The score is real; the licence is blocked. Blockage is an absence, not a weak licence.

Between these lies **VITIATED GENERATION**: something forms, but less than the report claims. Partial contamination, an unvalidated automated judge, and unmeasured prompt sensitivity each leave a licence standing that's narrower or weaker than its advertised scope. Many disputes about whether a benchmark "really measures" a construct are disputes about vitiation, and they go better when put that way, because vitiation turns on specifiable conditions rather than on essence.

A licence used beyond its scope isn't a weaker licence; for the wider claim, it's no licence at all. A source-faithfulness score in retrieval-augmented generation licenses claims about that corpus and retrieval design, not about open-domain factual accuracy. Exceeding scope is the characteristic misstep that the misrecognition axis below will name.

4.2 THE CAREER OF A LIVE LICENCE

Five things can become of a licence once generated.

New evidence can **NARROW** it. When controlled variants show that performance drops under value substitutions and irrelevant clauses, the original licence isn't cancelled; it persists with reduced scope, covering the original items and format but no longer the capability reading (Mirzadeh et al., 2025). A narrowed licence re-enters the cycle, where it can be narrowed again, defended, or given up.

Optimization against the measure **ERODES** it. A benchmark licence is issued under a tacit no-targeting condition: the score means what it means partly because no one was steering toward it. Reward-model overoptimization gives the general form of the problem, since optimizing a learned proxy outruns the target it stood in for (Gao et al., 2023), and leaderboard culture applies the same pressure to benchmarks. The familiar slogan that a measure which becomes a target stops being a good measure has a licence-theoretic reading: targeting doesn't falsify the score, it spends the licence. Saturation is erosion's endpoint.

The issuing conditions can lapse, and the licence EXPIRES. Models are updated, data distributions move, policies change. An evaluation of last year’s model licenses nothing about this year’s release, but the number survives in citation graphs and marketing long after the configuration it measured has gone. Expiry is undramatic, which is why it’s the easiest stage to miss.

A stronger result can DEFEAT it without invalidating it. A red-team success can defeat a deployment licence while the benchmark result it defeated still stands as an accurate record of what it measured. Defeat is override, not refutation, and the difference matters: a defeated licence can return if the defeating result is itself narrowed or expires.

The issuing practice can REVOKE it. Benchmarks are retracted when contamination is discovered, automated judges are replaced when bias is found, datasets are withdrawn. Revocation is uptake-facing: it tells the community to stop inferring, which is different from a licence quietly expiring.

4.3 MISRECOGNITION AND MAINTENANCE

One axis cuts across every stage. Each transition can be tracked, missed, or mistaken by the community that uses the licence. Uptake can treat a vitiated licence as full, an expired licence as live, a narrowed licence at its original scope. Model cards and surveys repeat claims whose supports have lapsed, and repetition doesn’t reissue a licence. MISRECOGNITION is second-order: a fact about the relation between a licence and its uptake, and explanatory in its own right. Benchmark essentialism, the habit of reading a benchmark as the capability itself, isn’t a one-off error but an entrenched misrecognition pattern with institutional supports: leaderboards reward it, citation practices spread it, and procurement shorthand depends on it.

The last piece is MAINTENANCE, where stability and control come apart. Repeated measurement can show that a score is stable; it can’t keep a licence current, any more than re-reading a thermometer maintains the temperature. Weinberger’s distinction between homeostasis and mere stability makes the point general: a system is controlled only when something corrects perturbations (Weinberger, 2026). An evaluation ecosystem maintains licences only where it has that control structure: monitoring that notices drift, re-evaluation under new items and attacks, release gates that respond to defeat, reissue after expiry. A leaderboard that never retests is a record, not a controller, and the licences it displays age at the speed of the field.

4.4 A DIAGNOSTIC TYPOLOGY

Single claims have careers. Across many claims, recurring term-roles show characteristic careers, and a compact typology is useful as a diagnostic. Table 1 classifies AI evaluation terms by the inference their evaluations are mainly used to license. The classification rule is a stipulation about primary role, not a discovery about usage statistics, and nothing below depends on a borderline assignment. Read each row from left to right: an observation becomes an operationalization, the operationalization licenses an inference, and the final columns state what can travel and which misrecognition the term characteristically invites.

Type	Observation	Operationalization	Licensed inference	Projectibility payoff	Characteristic misrecognition	Use context
Output-centred failure profile	generated output	factuality, faithfulness, support, uncertainty, granularity	the failure is likely to recur under specified structural supports	predicts recurrence, detection, and mitigation transfer under similar generation or grounding conditions	treating HALLUCINATION as false output, or treating one low score as deployment safety	model evaluation, retrieval-augmented generation, factuality claims
Adversarial relation	boundary-crossing interaction	attack distribution, policy class, evaluator, target system	the system is vulnerable under a specified adversarial network	predicts vulnerability and defence brittleness relative to attacker class, interface, and policy boundary	treating JAILBREAK RESISTANCE as a model property or projecting across attacker classes	red-teaming, release gates, security evaluation
Capability-attribution term	task performance	benchmark, metric, item population, prompting scaffold	the performance supports a capability claim over a domain	predicts transfer or robustness across item variants, metrics, scaffolds, and system boundaries when the attribution is validated	treating a score as a capability, or a benchmark as the domain	benchmark reporting, model comparison, deployment claims
Umbrella governance construct	aggregate risk or value label	component taxonomy, governance use	lower-level constructs are being coordinated for a purpose	predicts which lower-level evaluations should travel together for a stated governance decision	treating ALIGNMENT or TRUSTWORTHINESS as a single behavioural construct with a single score	procurement, audit, release and risk-management decisions

Table 1: AI evaluation terms by primary inferential role.

The typology is non-exclusive. Hallucination has relational features because unsupportedness depends on a source, a task, and a user context; jailbreaks often end in bad outputs; agency can be described as a capability, a system architecture, or a deployment relation. A term’s row records its primary role, nothing more.

Confused debates arise when one row’s licence is read with another row’s scope: a score read as a capability, a relational vulnerability read as a model property, an umbrella verdict read as a single behavioural cluster. The worked cases make the diagnostic concrete, and each also displays a different part of the life-cycle. Hallucination shows structure and scope; capability attribution shows generation, narrowing, and erosion; jailbreak shows expiry and defeat; and the governance verdicts show what uptake looks like when regimes rather than researchers hold the licences.

5 HALLUCINATION: THE KIND BEHIND THE LICENCE

HALLUCINATION starts from a generated output, but the failure profile is wider than falsity. It includes factuality, faithfulness to source, support and grounding, uncertainty, granularity, retrieval conditions, task context, and user uptake. The output remains the centre of the evaluation, but the source, retrieval system, prompt, and use context help determine which failure profile the output instantiates (L. Huang et al., 2025; Ji et al., 2023).

The source relation is central. In source-conditioned generation, an output can contradict the source, add claims that the source doesn’t verify, or introduce content that’s true but unsupported by

the assigned evidence. In open-domain question answering, the relevant source can be world knowledge rather than an input document. In retrieval-augmented generation, it can be the retrieved context, the retrieval corpus, or a cited passage. These settings share a family resemblance, but they don't license the same inference.

False output, then, is too coarse. A model can make a factual error, overstate what the available source warrants, contradict a retrieved document, fail to retrieve the needed evidence, answer with misplaced confidence under uncertainty, or give an answer whose harm depends on the user's task. These are related failures, but their operationalizations differ. A factuality metric for biographical claims doesn't measure source faithfulness in summarization, and a source-faithfulness score in retrieval-augmented generation doesn't measure whether a medical or legal assistant should have abstained.

Task goal matters too. If a system is asked to write a true-or-false test and includes "Toronto is the capital of Canada" as an item, the false proposition is intentional test content, not a hallucination. The same sentence becomes a hallucination if the system presents it as true, treats it as supported, or uses it as an answer where accuracy is the goal. The label tracks the role the output is meant to play, not falsity in the string alone.

The failure profile has recurring parts: a plausible generated answer, an evidential or factual defect, some relation to a source or task goal, and a user-facing risk created by the answer's uptake. Different evaluations put weight on different parts of the profile. A biography generator invites atomic factual checks; a summarizer invites source-faithfulness checks; a high-stakes assistant invites abstention and uncertainty checks.

So far, that's a list, and a list can't say which projections the classification licenses. The parts of the profile are ordered: some generate others, and the ordering is where the licence gets its content (L. Huang et al., 2025; Ji et al., 2023). Gaps in parametric knowledge generate the factual-error profile. Decoding and sampling choices generate the fluency and stated confidence that mask unsupportedness and give the failure its characteristic plausibility. Training and evaluation regimes that reward answering over abstaining generate the calibration profile, converting uncertainty into confident assertion. Retrieval design generates the source-faithfulness profile, because it fixes which evidence the output can be unfaithful to (Shuster et al., 2021). In Khalidi's terms, the profile is a node with internal causal structure, not a checklist (Khalidi, 2018).

That ordering explains the shape of the measurement literature. FActScore decomposes long-form answers into atomic facts and asks whether a specified source supports them: it reads the source relation (Min et al., 2023). SelfCheckGPT treats instability across sampled responses as a black-box signal of unsupported content: it reads the decoding position (Manakul et al., 2023). Retrieval augmentation intervenes on the evidence boundary rather than reading it, and can introduce noisy, missing, or conflicting context of its own (L. Huang et al., 2025; Shuster et al., 2021). Uncertainty-based methods read the calibration position: under some conditional-generation settings, higher predictive uncertainty tracks greater hallucination risk (Xiao & Wang, 2021).

Each operationalization licenses projection along the part of the structure it reads, and fails where its position predicts. A resampling detector, Section 3's simple case, misses the model that repeats the same false claim in every sample, because the decoding signal has disappeared. An atomic-fact score misses relevance, recall, and whether the system should have answered at all, because those sit away from the source relation. A retrieval intervention helps where missing parametric knowledge was the

bottleneck and adds failure modes of its own where it wasn't. These boundaries aren't peripheral complications. They're the licence's scope conditions, read off the kind's structure.

Read as a licence, then, a hallucination classification supports projection to recurrence and mitigation transfer where the generating structure is shared: the same knowledge boundary, decoding procedure, retrieval design, and incentive profile. Change one generator and the projection weakens exactly there, which is what a structured kind predicts and a flat label doesn't. And no position in the structure licenses the jump from one low hallucination score to deployment safety. That inference needs the coordination machinery of Section 8, stated rather than assumed.

6 CAPABILITY LICENCES: REASONING, EMERGENCE, AND AGENCY

REASONING, EMERGENCE, and AGENCY share a score-to-capability problem. Each begins with task performance and asks whether that performance supports a claim about what the system can do across a domain. The failure routes differ. Reasoning claims can be fragile across item variants, distractors, scaffolds, and contamination controls. Emergence claims can depend on metric choice. Agency claims can depend on where the model-tool-environment boundary is drawn.

The common risk is treating the benchmark as if it directly revealed the capability. A task score is an observation, a benchmark aggregate is an operationalization, and a capability attribution is an interpretation. The attribution becomes projectible only when the benchmark is embedded in a wider network of evidence: item variation, tests that separate the target construct from nearby constructs, contamination checks, prompting conditions, and comparisons with neighbouring constructs (Freiesleben, 2026).

In life-cycle terms, the recurring troubles sort by stage. Contamination is a generation failure: the capability licence never forms, however high the score. Unmeasured prompt sensitivity vitiates generation. Scaffolds set scope: if chain-of-thought prompting supplies the deliberation, the licence covers scaffolded performance, not unaided problem solving. And controlled item variation narrows licences after issue. Contamination checks are generation conditions, not optional extras.

Reasoning is the central case. The survey literature treats reasoning as a plural and unsettled construct, covering deductive, inductive, abductive, analogical, causal, probabilistic, formal, and informal reasoning (J. Huang & Chang, 2023). That plurality matters for validity. A benchmark can show strong performance on one task population while leaving open whether the model has a reasoning capacity that transfers across problem forms, domains, and prompting regimes.

The plurality also changes what counts as confirming evidence. A deductive reasoning claim asks for validity-preserving steps across form changes. A causal reasoning claim asks for sensitivity to interventions and counterfactual structure. A mathematical word-problem claim asks for stable mapping from the story to the relevant operations. Treating these as one flat reasoning score makes the construct easier to report and harder to validate.

GSM-Symbolic shows narrowing in action. Mirzadeh et al. (2025) generate controlled variants of GSM8K-style problems and show that model performance can vary substantially across value substitutions, problem complexity, and seemingly irrelevant clauses. The original licences aren't cancelled; they're narrowed, still covering the original items under the original format but no longer the general capability reading the reports had given them. A reasoning attribution grows stronger the more narrowing it survives: controlled variation, contamination checks, distractors, and changes of scaffold.

Emergence shows the apparatus situating a dispute rather than adjudicating it. Wei et al. (2022)

issue capability licences on an operational condition: an ability present in larger models and absent in smaller ones. Schaeffer et al. (2023) argue that under nonlinear or discontinuous metrics the generation conditions were never met: a sharp bend in the curve can report the metric’s geometry rather than a new capability. Put this way, the disagreement stops being a clash over the word *emergence* and becomes a definite question about generation: which invariances, across metrics, item populations, and model families, have to hold before a capability licence forms, and which are merely desirable. The apparatus doesn’t pick a side. It locates the fault-line, and a framework that can situate a dispute does more than one that can only label its parties.

The constructive lesson is to report the metric and task structure that make an emergence claim projectible. A sharp threshold under one scoring rule can license a claim about performance under that rule. It licenses a broader capability claim only when neighbouring metrics, item populations, and model families support the same interpretation.

Benchmark infrastructure then reads as licence maintenance. MMLU’s breadth widens scope, but its lopsided per-subject performance and calibration problems warn against reading the aggregate at face value (Hendrycks et al., 2020). HELM widens the generation conditions, requiring calibration, robustness, fairness, bias, toxicity, and efficiency alongside accuracy before broad licences issue (Liang et al., 2023). Dynabench is reissue machinery: when static benchmarks saturate, which is erosion’s endpoint, human-and-model-in-the-loop collection generates new items and, with them, new licences (Kiela et al., 2021).

Agency is the system-boundary case, where everything turns on scope. Agentic LLM surveys define the target as a system that receives natural-language input, reasons or plans, acts autonomously, uses tools or environments, and pursues goals (Plaat et al., 2025). The attribution attaches to a model-system configuration, not to a bare language model. Tool access, memory, planning horizon, feedback, permissions, and environment design help determine whether the system counts as agentic for a given evaluation.

The same base model can be part of different agency profiles. In an autocomplete interface, it predicts text without acting in the world. In a tool-using assistant, it can call APIs, revise plans, and affect an external state. In a multi-agent simulation, its behaviour depends on other agents and feedback from the environment. The agency licence travels only with the relevant system boundary.

AGI is the limiting case of a capability attribution. I’ve argued elsewhere that AGI evaluation goes better when general intelligence is treated as a property cluster rather than a checklist of tasks (Reynolds, 2025). In the present terms, an AGI attribution is a capability licence whose scope is contested all the way down, which is why no single benchmark could generate it.

The question “Does the model reason?” then dissolves into stages. What licence was generated, over which item population and scaffold? How far has it since been narrowed? How much erosion has leaderboard pressure inflicted? A capability term earns its keep when the answers predict transfer, robustness, and failure boundaries better than the raw score alone.

7 JAILBREAK: LICENCES UNDER AN ADAPTIVE ADVERSARY

JAILBREAK starts from an interaction that crosses a policy boundary. The evaluation object is a relation among model behaviour, attacker strategy, interface affordances, guardrail design, evaluator judgement, and the institutional definition of prohibited output. That’s why jailbreak resistance should be paired with an adversary model rather than treated as an intrinsic property of a

model (Vassilev et al., 2024; Yi et al., 2024). Direct chat jailbreaks, transferable adversarial suffixes, persuasion-based attacks, and indirect prompt injection share enough of this relational profile to be grouped together, but they don't share one mechanism (Greshake et al., 2023; Zeng et al., 2024; Zou et al., 2023).

This relational structure explains why in-the-wild jailbreaks are a social and technical population. Prompt families are copied, revised, hosted, tested, and patched across time. Their success depends on vendor policy, model update, interface, forbidden-content class, and evaluator choice (Shen et al., 2024). A benchmark that reports attack success rate without this network can still be useful, but the reported number has a narrow inferential target.

What distinguishes this row of the typology is tempo. Capability licences erode slowly; jailbreak licences date fast, because one of their scope conditions, the attack distribution, is an adversary that adapts on purpose. A resistance claim can be defeated by the next optimizer run and expires with the next interface or policy change. That's why red-teaming can't be a pre-release ritual. In life-cycle terms it's maintenance: the control loop that reissues or revokes resistance licences as the attack population moves (Weinberger, 2026). A safety case that cites last quarter's attack suite is running on an expired licence.

Classifying jailbreak as an adversarial relation licenses projection only relative to a specified adversarial network. A result can support expectations about vulnerability under attacker class A, interface I, policy boundary P, and attack distribution D. It doesn't by itself support expectations about attacker class B, a different interface, or a new policy regime. The payoff is a more precise release claim: resistance is always resistance to a defined attack space, for as long as that space stays the relevant one.

8 ALIGNMENT AND TRUSTWORTHINESS: THIN VERDICTS AND COORDINATION REGIMES

ALIGNMENT and TRUSTWORTHINESS differ from the other rows in role, not just in breadth. The typology called them umbrella constructs, naming the bundle of lower-level evaluations they manage. Seen from the label's side, they're THIN VERDICTS: they mark the outcome of an assessment without giving the route through the evaluations that produced it, the way *wrong* marks a moral conclusion without saying what was wrong with the act. A thin verdict projects only after some regime has fixed which lower-level evaluations feed it and how. Treating these terms as first-order behavioural kinds misfires: the projectible structure sits in the regime, not in the behaviour.

NIST's AI Risk Management Framework is best read in exactly that way, as a constitution for a coordination regime. It bundles validity and reliability, safety, security and resilience, accountability and transparency, explainability and interpretability, privacy, and fairness with harmful bias managed into TRUSTWORTHY AI, for the stated purposes of risk management (National Institute of Standards and Technology, 2023). The bundle is the content.

A COORDINATION REGIME, as I'll use the term, is the institutional structure that takes evaluations up: its assignment rules (which results count, at what thresholds), its records, its sanctions, and its decision points. A release gate, a procurement rule, an audit, and a research leaderboard are different regimes. The same lower-level result projects differently under each, because each fixes a different bundle and a different decision. A release decision and a procurement decision can use the same umbrella word while licensing different projections. That's regime-relativity, not equivocation, and it becomes equivocation only when the regime is left unstated.

The projectible claim is coordinative rather than behavioural. A TRUSTWORTHINESS label pre-

dicts that a specified set of lower-level evaluations has been assembled for a decision, that the same bundle can be inspected again in a similar use, and that omitting one component changes the governance claim. In life-cycle terms, the umbrella licence is only as current as the stalest licence in its bundle: a trustworthiness verdict resting on an expired robustness evaluation has expired too, whether or not anyone has noticed.

ALIGNMENT is similarly target-dependent. One target is user preference: RLHF trains models to follow instructions by optimizing against preference data and reward models (Ouyang et al., 2022). Another target is truthfulness. A third is policy compliance. A fourth is longer-term user welfare or institutional risk management. These targets can reinforce one another, but the construct claim changes when the target changes.

Consider a deployed assistant that gives agreeable answers to user beliefs, corrects false factual claims, and refuses prohibited requests under a content policy. A preference evaluation, a truthfulness evaluation, and a policy-compliance evaluation can assign different scores to the same system. Calling the system aligned says little until the lower-level target is stated.

Sycophancy is the pressure case. A behaviour can satisfy one alignment target, such as immediate user preference, while failing another, such as truthfulness. The same overoptimization that erodes benchmark licences (Section 4) operates here inside the reward signal: optimizing a learned proxy outruns the target it stood in for (Gao et al., 2023). Sycophancy research makes the conflict concrete: preference data and preference models can reward responses that match user beliefs over truthful corrections (Sharma et al., 2024), and recent formal work argues that preference-based post-training can amplify this behaviour when the learned reward favours agreement (Shapira et al., 2026).

The pressure case matters because it can reverse the evaluative verdict without changing the surface behaviour. Agreement with a false user belief can be a success relative to a learned preference signal and a failure relative to truthfulness, calibration, or epistemic support. The choice of lower-level target determines which inference is licensed, and the thin verdict shouldn't hide that choice.

The question "Is sycophancy misalignment?" is indexed to the lower-level construct in view. Relative to a narrow preference target, some sycophantic behaviour can be a local alignment success. Relative to truthfulness, calibration, user autonomy, or epistemic support, it's a failure. Only a regime that states its lower-level targets issues a determinate licence; an alignment claim should say which target is in view and block projection to incompatible targets. Without that specification, the thin verdict makes a decision sound measured while leaving the measured construct underdescribed.

9 OBJECTIONS AND LIMITS

Three objections deserve answers, and two limits deserve concession.

The first objection says the licence vocabulary is metaphor. Licences are deontic items, issued and revoked by authorities; evaluation warrants are epistemic, and nobody signs them. The framework borrows the vocabulary for its structure, not its metaphysics. Every stage names an event evaluation practice already exhibits without the vocabulary: benchmarks are retracted, results are patched around, claims outlive the configurations they measured, model cards repeat lapsed numbers. What the deontic idiom adds is articulation: separate names for failures that *invalid* runs together, and a distinction (licence versus uptake) that the idiom of truth and evidence doesn't draw by itself. A reader who wants the idiom cashed out can read *licence* as *warranted inference under stated conditions* throughout; nothing in the worked cases changes.

The second objection says the framework proves too much: any evaluation dispute can be re-described as some stage of some licence's career, so the apparatus explains everything and forbids nothing. But the account has failure conditions. It would fail if evaluation claims that travel well systematically lacked identifiable generation conditions and structural supports, since then projectibility wouldn't track structure. It would fail if stage assignments could be reshuffled case by case to save a diagnosis, since then the stages would carry no information. And it would fail if licence and uptake never came apart, since then the framework's central distinction would idle. The worked cases put content into each condition: contamination is a generation fact, GSM-Symbolic narrowing is an evidential fact, and benchmark essentialism is an observed gap between what scores license and how they're used.

The third objection says the framework is external validity renamed. The reply compresses Section 3. External validity and transportability ask whether a claim holds elsewhere, taking the variables and categories as given (Bareinboim & Pearl, 2016). The present question is prior, asking which classifications deserve projection at all, and wider, asking what generates, maintains, and revokes the entitlement to project, and who misreads it. A theory of truth-travel has no stage called revocation and no axis called misrecognition; evaluation practice visibly has both.

Two limits should be conceded. The paper works two cases in depth and places the others: jailbreak and the governance verdicts get sketches whose details would need their own papers. And the framework classifies rather than adjudicates. Locating the emergence dispute at generation doesn't settle it, and mapping a licence regime doesn't certify the decisions the regime serves; a well-maintained licence architecture can still serve a bad policy. Explanation and vindication are different achievements, and the framework offers the first.

What remains is the contribution. Goodman supplies the term, Boyd and Khalidi the structure-first account of projection, Kane the argument-based view of validation, Weinberger the difference between stability and control. None of them supplies the life-cycle: generation, scope, narrowing, erosion, expiry, defeat, revocation, misrecognition, maintenance. The materials are borrowed; the life-cycle is not.

10 CONCLUSION

AI evaluation has explicit norms for reliability and validity, and mostly habits for travel. The proposal here is that the missing norm has a structure. Evaluation terms function as inference licences, particular evaluations issue them, and the issued licences have life-cycles: each stage carries stateable conditions, with misrecognition and maintenance as the cross-cutting axes. The typology of term-roles then works as a diagnostic: output-centred failure profiles, adversarial relations, capability attributions, and thin governance verdicts fail in characteristically different ways, and disputes go wrong when one role's licence is read with another's scope.

The worked cases show what the apparatus buys. Hallucination detectors turn out to read different positions in one generating structure, which fixes where each detector's licence stops. The emergence dispute relocates to a definite question about generation conditions rather than a quarrel over a word. Jailbreak resistance dates as fast as its adversary adapts, which makes red-teaming maintenance rather than ritual. And an umbrella verdict is only as current as the stalest licence in its bundle.

The framework has a practical corollary. An evaluation report could carry a licence statement:

the claim licensed, its scope, the structural supports it rests on, the known breakers, and the conditions under which it should be treated as expired. Model and system cards already have slots this information would fit. Most of it exists at evaluation time and is lost by reporting conventions built around single numbers; a licence statement is a format for keeping it.

The typology is a starting point rather than a closed map. BIAS is the obvious extension. It looks partly like an output profile, partly like a deployment-sensitive relation among model, population, task, and affected group, and partly like a governance construct, so its licences will have an unusually layered career. The useful question stays the same throughout: what does the term license in a given evaluation, what structure lets the inference travel, what would spend or defeat it, and who's checking that it's still alive?

ACKNOWLEDGEMENTS

The large language models OpenAI Codex, GPT-5.5 Pro, and Claude Fable 5 served as drafting and editing aids throughout the preparation of this paper. I am responsible for all theoretical claims, arguments, errors, and interpretive choices.

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